

Section: 360UB44.7 Span: 10.00 m

FAIL

Job No: J001
Job Name: Town Hall

Inputs

Span: 10.00 m
Tributary width: 3.00 m
Restraint: advanced, advanced F-F
Loads:
 L: 20.00 kN/m, 0.00-10.00 m (G)

Section Properties

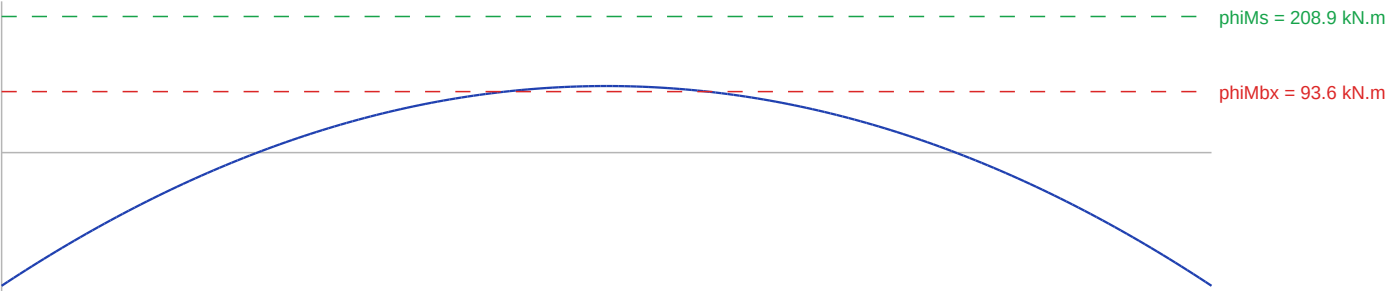
Designation	360UB44.7
Mass	44.7 kg/m
d	352.0 mm
bf	171.0 mm
tf	9.70 mm
tw	6.90 mm
Ix	1.21e+8 mm^4
Sx	7.77e+5 mm^3
Zx	6.89e+5 mm^3
Iy	8.10e+6 mm^4
J	1.19e+5 mm^4
Iw	2.37e+11 mm^6
fy	300 MPa

Design Check Summary

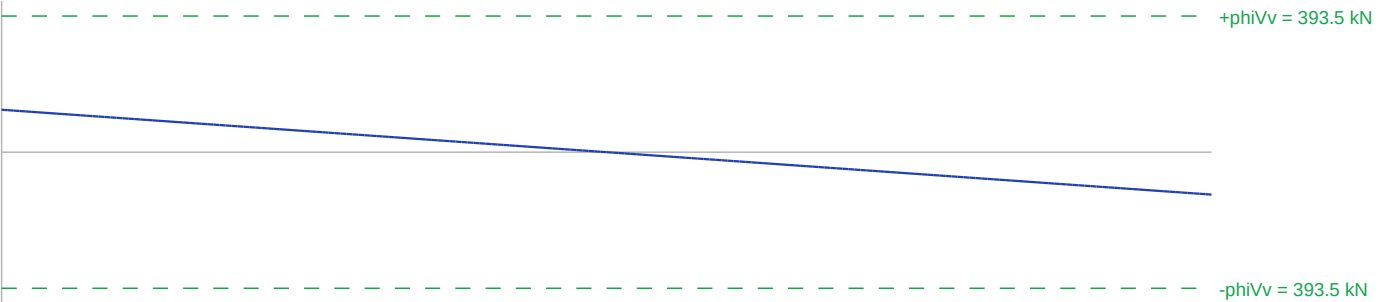
Check	Demand	Capacity	Util	Status
Section moment	204.4 kN.m	208.9 kN.m	97.8%	PASS
Member moment	204.4 kN.m	93.6 kN.m	218.3%	FAIL
Shear	122.6 kN	393.5 kN	31.2%	PASS
Deflection G+0.4Q	22.0 mm	33.3 mm	66.0%	PASS
Deflection G	22.0 mm	27.8 mm	79.2%	PASS

Diagrams

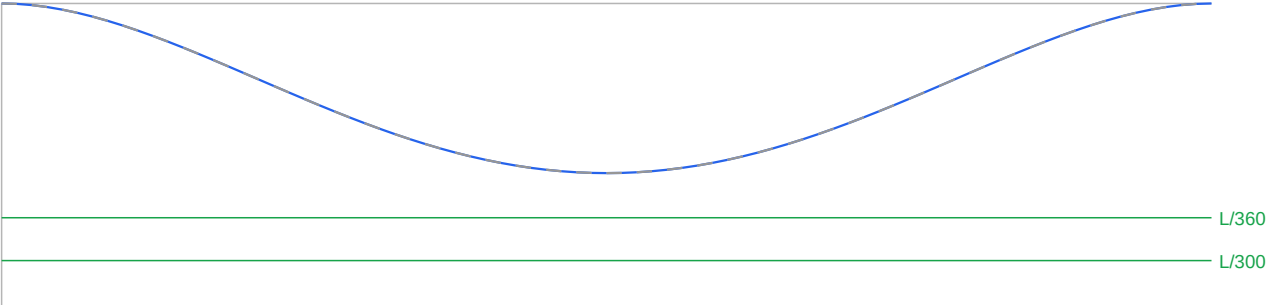
Bending Moment Diagram (kN.m)



Shear Force Diagram (kN)



Deflection Profile (mm)



1. Material Properties

[AS4100 Cl. 2.1]

 $f_y = 300 \text{ MPa}$ ($t_f = 9.7 \text{ mm}$)**2. Section Classification - Flange**

[AS4100 Cl. 5.2.2]

 $\lambda_{f} = (b_f / 2t_f) \times \sqrt{f_y/250} = 9.27$ Compact limit $\lambda_{ep} = 9 \rightarrow \lambda_f > \lambda_{ep} \rightarrow$ Flange: Non-compact**3. Section Classification - Web**

[AS4100 Cl. 5.2.2]

 $\lambda_w = (d - 2t_f)/t_w \times \sqrt{f_y/250} = 52.80$ Compact limit $\lambda_{ep} = 82 \rightarrow \lambda_w \leq \lambda_{ep} \rightarrow$ Web: Compact

Overall section class: Non-compact

4. Effective Section Modulus

[AS4100 Cl. 5.2.3]

 $Z_e = 7.74e+5 \text{ mm}^3$ (per Non-compact section)**5. Section Moment Capacity**

[AS4100 Cl. 5.1]

 $M_{sx} = Z_e \times f_y = 7.74e+5 \times 300 = 2.32e+8 \text{ N.mm}$ $\phi M_s = 0.9 \times M_{sx} / 1e6 = 208.9 \text{ kN.m}$ **6. Effective Length & Supports**

[AS4100 Cl. 5.6.3]

Support: Fixed-Fixed

 $L_e = 10.00 \text{ m}$ End moments (1.2G+1.5Q): $M_A = 204.4 \text{ kN.m}$, $M_B = 204.4 \text{ kN.m}$ (hogging)**7. Reference Buckling Moment**

[AS4100 Cl. 5.6.1.1]

 $M_{oa} = (\pi/L_e) \times \sqrt{E \cdot I_y \times (G \cdot J + \pi^2 \cdot E \cdot I_w / L_e^2)} = 47.6 \text{ kN.m}$ **8. Member Slenderness Reduction**

[AS4100 Cl. 5.6.1]

 $\alpha_m = 2.50$ (moment distribution factor) $\alpha_s = 0.6 \times (\sqrt{(M_{sx}/M_{oa})^2 + 3}) - M_{sx}/M_{oa} = 0.179$ **9. Member Moment Capacity**

[AS4100 Cl. 5.6]

 $\phi M_{bx} = \min(0.9 \times \alpha_m \times \alpha_s \times M_{sx}, \phi M_s) = 93.6 \text{ kN.m}$ **10. Shear Capacity**

[AS4100 Cl. 5.11.4]

 $A_w = 2429 \text{ mm}^2$ $d/t_w = 51.0 \leq 82 \cdot \sqrt{250/f_y} = 74.9 \rightarrow$ no web slenderness reduction $\phi V_v = 0.9 \times 0.6 \times f_y \times A_w / 1000 = 393.5 \text{ kN}$ **COMBINED ACTIONS**

[AS4100 Cl. 8.4]

 $N^* = 120.0 \text{ kN}$ $\phi N_s = 0.9 \times A_g \times f_y = 0.9 \times 5720 \times 300 / 1000 = 1544.4 \text{ kN}$ $\lambda_n = (L_e/r_{min}) \times \sqrt{k_f \times f_y / 250} = 291.1$ ($k_f = 1.00$) $\alpha_c = 0.087$ (HR residual stress category, Table 6.3.3(a)) $\phi N_c = 0.9 \times \alpha_c \times k_f \times A_g \times f_y / 1000 = 134.8 \text{ kN}$ Section: $N^*/\phi N_s + M^*/\phi M_s = 1.056 \leq 1.0 \rightarrow$ FAIL [Cl. 8.4.2.1]Member: $N^*/\phi N_c + M^*/\phi M_{bx} = 3.073 \leq 1.0 \rightarrow$ FAIL [Cl. 8.4.2.2]**11. Load Combinations**

[AS1170.0 Cl. 4.2.2]

ULS governing: 1.2G+1.5Q $\rightarrow M_{max} = 204.4 \text{ kN.m}$, $V_{max} = 122.6 \text{ kN}$ SLS combo: G + ψ_I x Q ($\psi_I = 0.4$, Office)**12. Deflection Check - G+ ψ_I .Q**

[AS1170.1 Cl. 4 / AS4100 Cl. 1.5.3]

Live load type: Office $\rightarrow \psi_I = 0.4$ $\delta = 22.0 \text{ mm} \leq L/300 = 33.3 \text{ mm} \rightarrow$ PASS

13. Deflection Check - G only

[AS4100 Cl. 1.5.3]

delta = 22.0 mm $\leq$ L/360 = 27.8 mm -> PASS
